

- 0: No delay.
- 1: Add 50ns delay. (Default value).

SDA_OUTPUT_DELAYED

SDA_OUTPUT_DELAYED attribute is used to add 50ns additional delay to the SDAO signal.

```
/* synthesis SDA_OUTPUT_DELAYED= value */
```

Value:

- 0: No delay (Default value).
- 1: Add 50ns delay.

SCL_INPUT_FILTERED

SCL_INPUT_FILTERED attribute is used to add 50ns glitch filter to the SCLI signal.

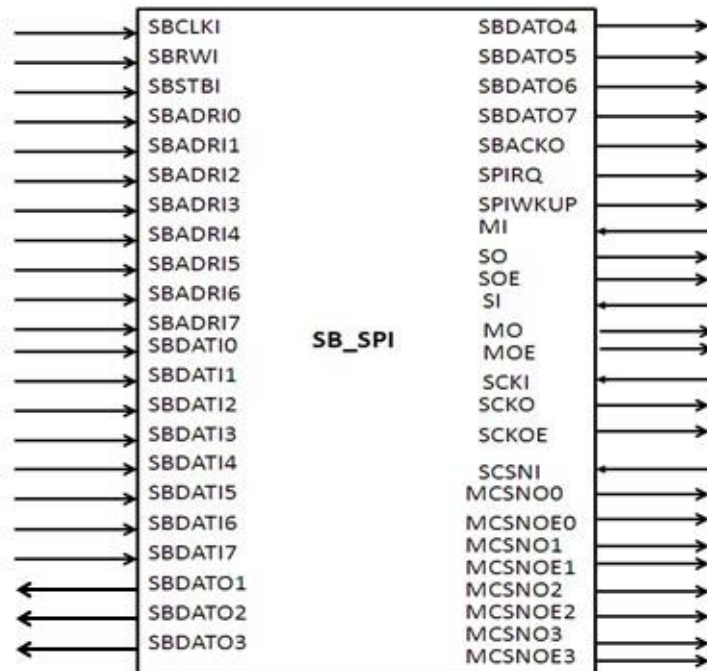
```
/* synthesis SCL_INPUT_FILTERED= value */
```

Value:

- 0: No Filter delay (Default value).
- 1: Add 50ns glitch filter delay.

SB_SPI

iCE40UP device supports two SPI hard IP primitives, located at lower left corner and lower right corner of the chip. The port interface of SB_SPI is similar as ICE40LM/iCE5LP SB_SPI primitive. Refer Page 117.



Ports

The port interface is similar as iCE40LM/iCE5LP SB_SPI primitive. Refer Page 117.

Parameters

The parameters are same as ICE40LM/iCE5LP SB_SPI primitive.

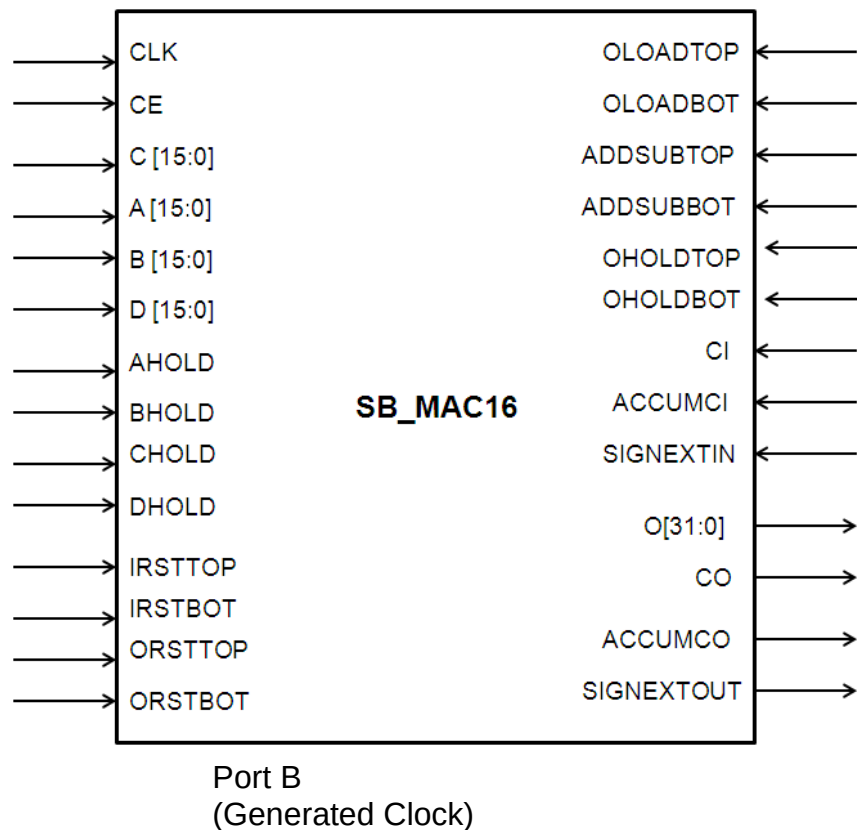
Synthesis Attribute

The synthesis attribute is same as ICE40LM/iCE5LP SB_SPI primitive.

SB_MAC16

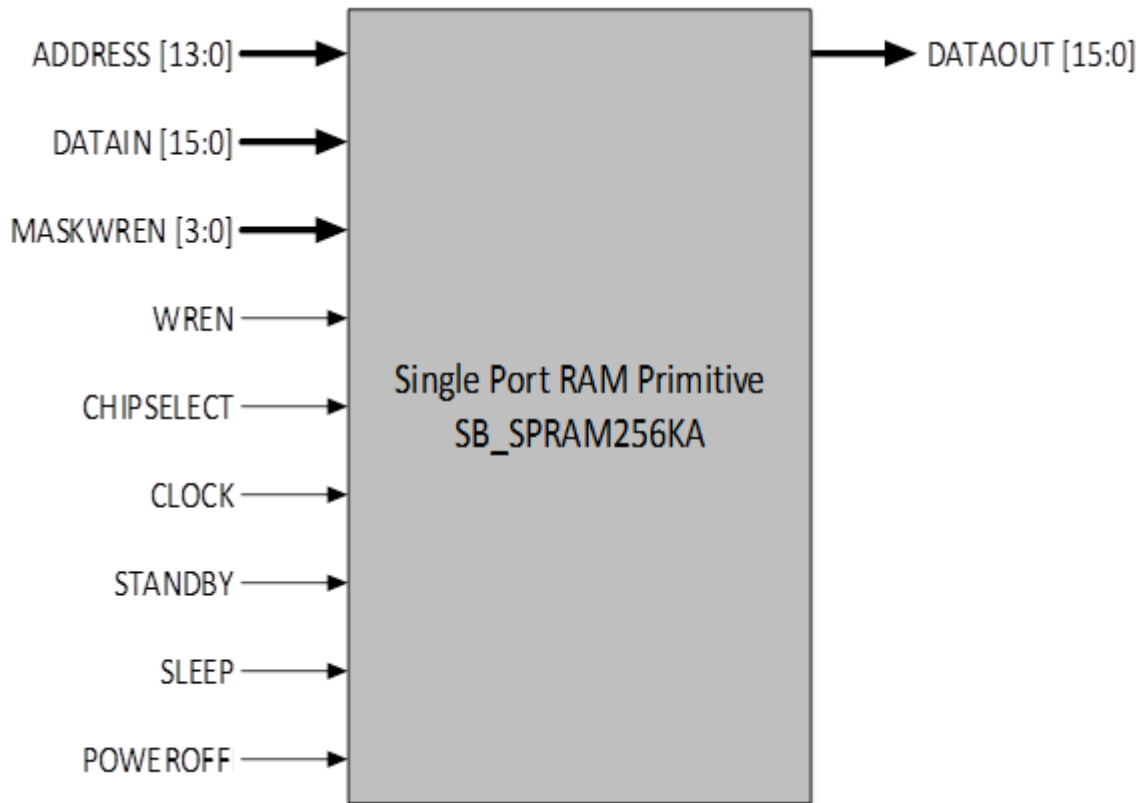
The SB_MAC16 primitive is the dedicated configurable DSP block available in iCE5LP/iCE40UP devices. The SB_MAC16 can be configured into a multiplier, adder, subtracter, accumulator, multiply-add and multiply-sub through the instance parameters. The SB_MAC16 blocks can be cascaded to implement wider functional units.

iCEcube2 supports a set of predefined SB_MAC16 functional configurations. Refer Page 133 for SB_MAC16 functional model, port details and the list of supported configurations.



SB_SPRAM256KA

iCE40UP devices offer four embedded memory blocks of Single Port RAM. Each of these blocks can be configured in 16K x 16 mode. These RAM blocks can be cascaded using logic implemented in the fabric to create larger memories.



Ports

Port Name	Port Width	Default Value	Description
ADDRESS	[13:0]	14b'000000 00000000	This Address input port is used to address the location to be written during the write cycle and read during the read cycle.
DATAIN	[15:0]	16b'000000 0000000000	The data input bus used to write the data into the memory location specified by address input port during the write cycle.

MASKWREN	[3:0]	4b'1111	The Bit Write feature where selective write to individual I/O's can be done using the Maskable Write Enable signals. Each bit masks a nibble of DATAIN.
WREN	[0:0]	1b'0	When the Write Enable input is Logic High, the memory is in the write cycle. When the Write enable is Logic Low, the memory is in the Read Cycle.
CHIPSELECT	[0:0]	1b'0	When the memory enable input is Logic High, the memory is enabled and read/write operations can be performed. When memory Enable input is logic Low, the memory is deactivated.
CLOCK	[0:0]	-	This is the external clock for the memory.
STANDBY	[0:0]	1b'0	When this pin is active then memory goes into low leakage mode, there is no change in the output state.
SLEEP	[0:0]	1b'0	This pin shut down power to periphery and maintain memory contents. The outputs of the memory are pulled low.
POWEROFF	[0:0]	1b'1	This pin turns off the power to the memory core when it is driven low (1'b0). There is no memory data retention when it is driven low. When POWEROFF is driven high (1'b1), the SPRAM is powered on.
DATAOUT	[15:0]	-	It outputs the contents of the memory location addressed by the Address Input signals.

The default value of each MASKWREN is 1. In order to mask a nibble of DATAIN, the MASKWREN needs to be pulled low (0).

The following shows which MASKWREN bits enables the mask for the DATAIN nibbles.

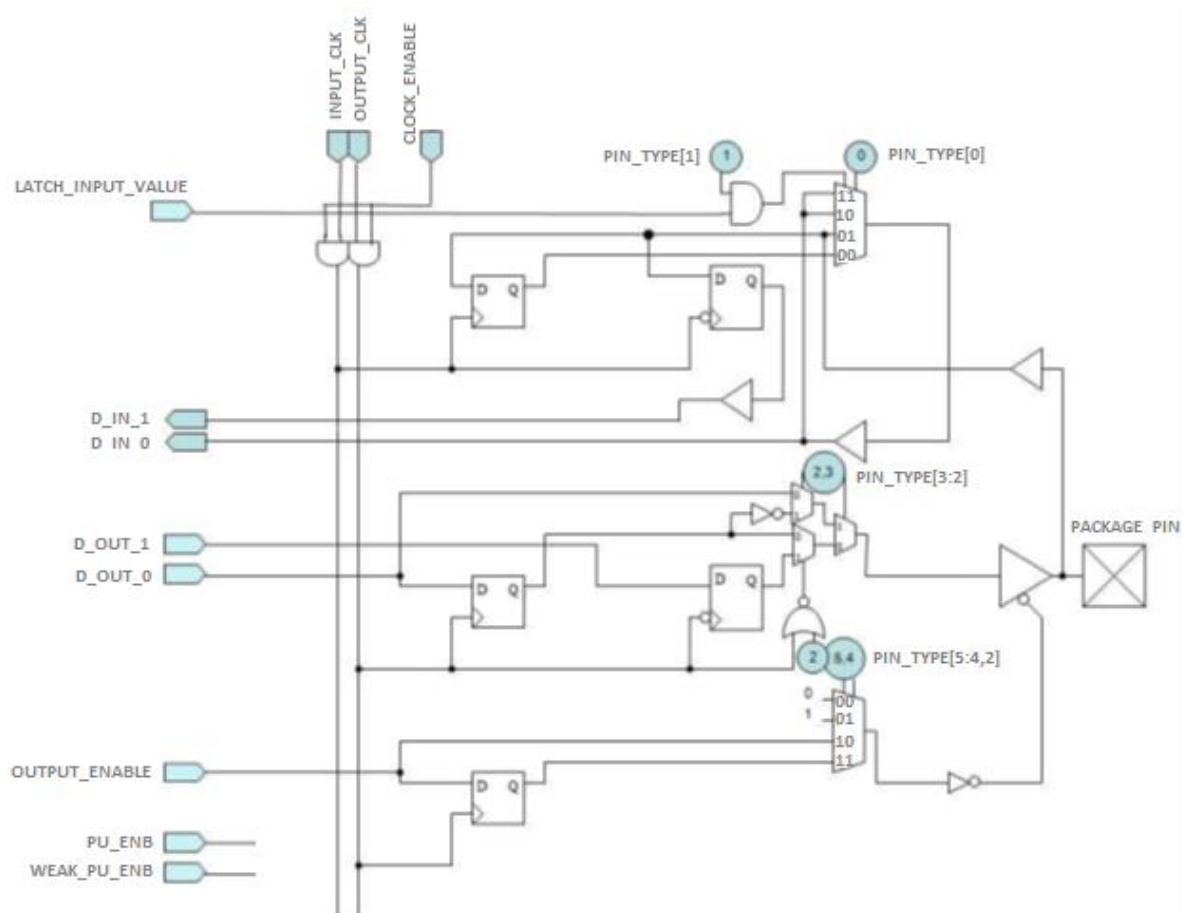
MASKWREN(3) enables mask for DATAIN(15:12)
 MASKWREN(2) enables mask for DATAIN (11:8)
 MASKWREN(1) enables mask for DATAIN (7:4)

MASKWREN(0) enables mask for DATAIN(3:0)

SB_IO_I3C

ICE40UP devices contain 2 special purpose IO blocks. These blocks are same as SB_IO, but have two additional user control signals PU_ENB, WEAK_PU_ENB to enable/disable the resistor networks. These blocks are useful in I3C applications.

The SB_IO_I3C block contains five registers. The following figure and Verilog template illustrate the complete user accessible logic diagram, and its Verilog instantiation.



Default Signal Values

The iCEcube2 software assigns the logic '0' value to all unconnected input ports except for CLOCK_ENABLE.

Note that explicitly connecting a logic '1' value to port CLOCK_ENABLE will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the Input and Output registers always enabled, it is recommended that port CLOCK_ENABLE be left unconnected.

Explicitly connecting a logic "0" value to port PU_ENB and/or WEAK_PU_ENB will result extra LUTs will be used to generate the Logic "0". If it is intended to always enable the pull-up and/or weak pull-up, it is recommended the associated port be left unconnected.

Input and Output Pin Function Tables

Input and Output functions are independently selectable via PIN_TYPE [1:0] and PIN_TYPE [5:2] parameter settings respectively. Specific IO functions are defined by the combination of both attributes. This means that the complete number of combinations is 64, although some combinations are not valid and not defined below.

Note that the selection of IO Standards such as LVCMOS etc is not defined by these tables.

Input Pin Function Table				
#	Pin Function Mnemonic	PIN_TYPE[1:0]		Functional Description of Package Pin Input Operation
1	PIN_INPUT	0	1	Simple input pin (D_IN_0)
2	PIN_INPUT_LATCH	1	1	Disables internal data changes on the physical input pin by latching the value.
3	PIN_INPUT_REGISTERED	0	0	Input data is registered in input cell
4	PIN_INPUT_REGISTERED_LATCH	1	0	Disables internal data changes on the physical input pin by latching the value on the input register
5	PIN_INPUT_DDR	0	0	Input 'DDR' data is clocked out on rising and falling clock edges. Use the D_IN_0 and D_IN_1 pins for DDR operation.

Output Pin Function table						
#	Pin Function Mnemonic	PIN_TYPE[5:2]				Functional Description of Package Pin Output Operation
1	PIN_NO_OUTPUT	0	0	0	0	Disables the output function
2	PIN_OUTPUT	0	1	1	0	Simple output pin, (no enable)
3	PIN_OUTPUT_TRISTATE	1	0	1	0	The output pin may be tristated using the enable
4	PIN_OUTPUT_ENABLE_REGISTERED	1	1	1	0	The output pin may be tristated using a registered enable signal
5	PIN_OUTPUT_REGISTERED	0	1	0	1	Output registered, (no enable)
6	PIN_OUTPUT_REGISTERED_ENABLE	1	0	0	1	Output registered with enable (enable is not registered)
7	PIN_OUTPUT_REGISTERED_ENABLE_REGISTERED	1	1	0	1	Output registered and enable registered
8	PIN_OUTPUT_DDR	0	1	0	0	Output 'DDR' data is clocked out on rising and falling clock edges
9	PIN_OUTPUT_DDR_ENABLE	1	0	0	0	Output data is clocked out on rising and falling clock edges
10	PIN_OUTPUT_DDR_ENABLE_REGISTERED	1	1	0	0	Output 'DDR' data with registered enable signal
11	PIN_OUTPUT_REGISTERED_INVERTED	0	1	1	1	Output registered signal is inverted
12	PIN_OUTPUT_REGISTERED_ENABLE_INVERTED	1	0	1	1	Output signal is registered and inverted, (no enable function)
13	PIN_OUTPUT_REGISTERED_ENABLE_REGISTERED_INVERTED	1	1	1	1	Output signal is registered and inverted, the enable/tristate control is registered.

Syntax Verilog Use

Output Pin Function is the bit vector associated with PIN_TYPE [5:2] and Input Pin Function is the bit vector associated with PIN_TYPE [1:0], resulting in a 6 bit value PIN_TYPE [5:0]

```
defparam my_generic_IO.PIN_TYPE = 6'b{Output Pin Function, Input Pin Function};
```

Pull Up Resistor Configuration.

The user can configure the internal pull up resistor strength of an IO to a predefined resistor value via parameter settings and synthesis attributes. The PULLUP parameter setting enables one of the resistor values as given in table 1 and WEAKPULLUP parameter setting enables the 100K pull up resistor.

The specific resistor value associated with the PULLUP parameter needs to specify via PULLUP_RESISTOR attribute. The PULLUP_RESISTOR value is effective only when PULLUP parameter is set to 1.

Both the PULLUP & WEAKPULLUP parameters are allowed to set simultaneously. When both parameters are set the effective pull up resistor values is [100K || {3P3K/6P8K/10K}]

Once the required parameters, attribute values are set, the active low PU_ENB, WEAK_PU_ENB signals allows the user to control the pull up resistor network dynamically.

Synthesis Attribute Syntax:

```
/* synthesis PULLUP_RESISTOR = "3P3K" */
```

Resistor Value:

Resistor Value	Description.
"3P3K"	Pull up resistor level is 3.3K.
"6P8K"	Pull up resistor level is 6.8K.
"10K"	Pull up resistor level is 10K.

Note: The PULLUP_RESISTOR value is effective only when PULLUP parameter is set to 1.

Verilog Instantiation

```
SB_IO_I3C    I3CIO_INST
(
  .PACKAGE_PIN (Package_Pin),           // User's Pin signal name
  .LATCH_INPUT_VALUE (latch_input_value), // Latches/holds the Input value
  .CLOCK_ENABLE (clock_enable),          // Clock Enable common to input and
                                          // output clock
  .INPUT_CLK (input_clk),                 // Clock for the input registers
  .OUTPUT_CLK (output_clk),               // Clock for the output registers
  .OUTPUT_ENABLE (output_enable),         // Output Pin Tristate/Enable
                                          // control
  .D_OUT_0 (d_out_0),                     // Data 0 - out to Pin/Rising clk
                                          // edge
  .D_OUT_1 (d_out_1),                     // Data 1 - out to Pin/Falling clk
                                          // edge
  .D_IN_0 (d_in_0),                       // Data 0 - Pin input/Rising clk
                                          // edge
  .D_IN_1 (d_in_1),                       // Data 1 - Pin input/Falling clk
                                          // edge
)
```



```

.PU_ENB (pu_enb),                                // Active low Pull Up resistor
                                                    Enable -Default to GND
.WEAK_PU_ENB (weak_pu_enb)                        //Active low Weak Pull Up Enable
                                                    for 100K resistor. Default to GND

) /* synthesis PULLUP_RESISTOR "3P3K" */ ;

defparam I3CIO_INST.PIN_TYPE = 6'b000000;
                                                    // See Input and Output Pin Function Tables.
                                                    // Default value of PIN_TYPE = 6'b000000 i.e.
                                                    // an input pad, with the input signal
                                                    // registered.
defparam I3CIO_INST.PULLUP = 1'b0;
                                                    // By default, the IO will have NO pull up.
defparam I3CIO_INST.WEAK_PULLUP = 1'b0;
                                                    // By default, the IO will have NO weak pull
up.
defparam I3CIO_INST.NEG_TRIGGER = 1'b0;
defparam I3CIO_INST.IO_STANDARD = "SB_LVCMOS";

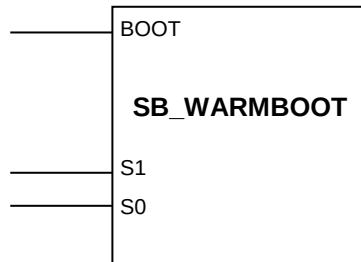
```

Device Configuration Primitives

SB_WARMBOOT

ICE FPGA devices permit the user to load a different configuration image during regular operation. Through the use of the Warm Boot Primitive, the user can load one of 4 pre-defined configuration images into the ICE FPGA device.

Note that this Warm Boot mode is different from the Cold Boot operation, which is executed during the initial device boot-up sequence.



The selection of one of these 4 images is accomplished through 2 input signals, S1 and S0. In order to trigger the selection of a new image, an additional signal, BOOT, is provided. It should be noted that this signal is level-triggered, and should be used for every Warm Boot operation i.e. every time the user wishes to load a new image into the device.

The successful instantiation of this primitive also requires the user to specify the address locations of the 4 images. These addresses should be specified in the iCEcube2 software as per the Warm Boot Application Note.

Verilog Instantiation

```
SB_WARMBOOT my_warmboot_i (
    .BOOT (my_boot),           // Level-sensitive trigger signal
    .S1 (my_sel1),             // S1, S0 specify selection of the
    .S0 (my_sel0)               // configuration image
);
```